

Borui Zhang

Postdoctoral Researcher, Tsinghua University

Co-founder & CEO, LoopEva

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[Google Scholar](#) | [GitHub](#) | [LoopEva](#)



About Me

- Postdoctoral researcher in Prof. Jiwen Lu's laboratory at Tsinghua University, leading research in the Embodied Intelligence Systems Group, and co-founder and CEO of LoopEva. My work connects visual representation learning, efficient multimodal models, explainable learning, and embodied intelligent systems.

Appointments

- Sep. 2026 - Present ■ **Postdoctoral Researcher · Shuimu Scholar**, Tsinghua University.
- Current ■ **Co-founder & CEO**, LoopEva.

Education

- Sep. 2021 - Jun. 2026 ■ **Ph.D. in Control Science and Engineering**, Tsinghua University · Department of Automation.
Advisor: Prof. Jiwen Lu.
- Sep. 2017 - Jun. 2021 ■ **B.E. in Automation**, Tsinghua University · Department of Automation.
GPA: 3.81/4.0. Ranked 1st out of 181 students.
- Sep. 2018 - Jun. 2021 ■ **B.A. in Economics · Second Degree**, Tsinghua University · School of Economics and Management.

Research Interests

- Embodied Intelligence & Multimodal Agents
- Efficient Models & Visual Representation
- Explainable Learning & Model Optimization

Research Publications

* indicates equal contribution. Borui Zhang's name is shown in bold.

Journal Articles

- 1 **Borui Zhang***, Baotong Tian*, Wenzhao Zheng, Jie Zhou, Jiwen Lu, "Exploring Unified Perspective For Fast Shapley Value Estimation," *Chinese Journal of Electronics*, 2026.
- 2 **Borui Zhang**, Qihang Rao, Jie Zhou, Jiwen Lu, "Alignment-Invertibility Regularization for Explainable Neural Networks," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2026.
- 3 Kangjie Liu, **Borui Zhang**, Jiwen Lu, Haibin Yan, "Toward Integrity and Detail With Ensemble Learning for Salient Object Detection in Optical Remote-Sensing Images," *IEEE Transactions on Geoscience and Remote Sensing*, 2024.

Conference Proceedings

- 1 **Borui Zhang**^{*}, Bo Zhang^{*}, Bo Wang^{*}, Wenzhao Zheng, Yuhao Cheng, Liang Tang, Yiqiang Yan, Jie Zhou, Jiwen Lu, "BAMI: Training-Free Bias Mitigation in GUI Grounding," *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.
- 2 Yuanhui Huang^{*}, Wenzhao Zheng^{*}, **Borui Zhang**, Jie Zhou, Jiwen Lu, "SelfOcc: Self-Supervised Vision-Based 3D Occupancy Prediction," *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 19946-19956.
- 3 **Borui Zhang**, Wenzhao Zheng, Jie Zhou, Jiwen Lu, "Path Choice Matters for Clear Attribution in Path Methods," *International Conference on Learning Representations (ICLR)*, 2024.
- 4 Wenzhao Zheng^{*}, Weiliang Chen^{*}, Yuanhui Huang, **Borui Zhang**, Yueqi Duan, Jiwen Lu, "OccWorld: Learning a 3D Occupancy World Model for Autonomous Driving," *European Conference on Computer Vision (ECCV)*, 2024.
- 5 Linqing Zhao, Xiwei Xu, Ziwei Wang, Yunpeng Zhang, **Borui Zhang**, Wenzhao Zheng, Dalong Du, Jie Zhou, Jiwen Lu, "LowRankOcc: Tensor Decomposition and Low-Rank Recovery for Vision-based 3D Semantic Occupancy Prediction," *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 9806-9815.
- 6 **Borui Zhang**, Wenzhao Zheng, Jie Zhou, Jiwen Lu, "Bort: Towards Explainable Neural Networks with Bounded Orthogonal Constraint," *International Conference on Learning Representations (ICLR)*, 2023.
- 7 Wenzhao Zheng, Yuanhui Huang, **Borui Zhang**, Jie Zhou, Jiwen Lu, "Dynamic Metric Learning with Cross-Level Concept Distillation," *European Conference on Computer Vision (ECCV)*, 2022, pp. 197-213.
- 8 **Borui Zhang**, Wenzhao Zheng, Jie Zhou, Jiwen Lu, "Attributable Visual Similarity Learning," *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
- 9 Wenzhao Zheng^{*}, **Borui Zhang**^{*}, Jiwen Lu, Jie Zhou, "Deep Relational Metric Learning," *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.

Preprints

- 1 Minglei Shi, Haolin Wang, **Borui Zhang**, Wenzhao Zheng, Bohan Zeng, Ziyang Yuan, Xiaoshi Wu, Yuanxing Zhang, Huan Yang, Xintao Wang, Pengfei Wan, Kun Gai, Jie Zhou, Jiwen Lu, "SVG-T2I: Scaling Up Text-to-Image Latent Diffusion Model Without Variational Autoencoder," *arXiv preprint*, 2025.
- 2 Qihang Rao^{*}, **Borui Zhang**^{*}, Wenzhao Zheng, Jie Zhou, Jiwen Lu, "SFTok: Bridging the Performance Gap in Discrete Tokenizers," *arXiv preprint*, 2025.
- 3 **Borui Zhang**^{*}, Qihang Rao^{*}, Wenzhao Zheng, Jie Zhou, Jiwen Lu, "Quantize-then-Rectify: Efficient VQ-VAE Training," *arXiv preprint*, 2025.
- 4 **Borui Zhang**, Wenzhao Zheng, Jie Zhou, Jiwen Lu, "Preventing Local Pitfalls in Vector Quantization via Optimal Transport," *arXiv preprint*, 2024.

Honors and Awards

2026	Shuimu Scholar, Tsinghua University
	Outstanding Graduate, Beijing
	Excellent Graduate, Tsinghua University
2025	Outstanding Graduate Student CPC Member, Tsinghua University
2024	12.9 Student Counselor Award, Tsinghua University
2023	National Scholarship
2022	Outstanding Student Cadre Award, Tsinghua University
	Excellent Teaching Assistant Award, Tsinghua University
2021	Future Scholars Scholarship, Tsinghua University (rank: 1/181)
2020	Changtong Scholarship
2019	Jiang Nanxiang Scholarship
2018	National Scholarship

Academic Services

- Conference reviewer.** CVPR (2022, 2024), ICCV (2023), ECCV, NeurIPS (2024), ICME (2022-2024), ACCV (2024), VCIP (2022)
- Journal reviewer.** IEEE Transactions on Image Processing

Teaching

2021-2022	Teaching Assistant · Numerical Analysis , Tsinghua University.
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Interests

- Photography, travel, classical music, badminton, table tennis, and running.